Quadratic Functions and Equations

Quadratic function

A **quadratic function** has the form

$$f(x) = ax^2 + bx + c, \qquad a \neq 0,$$

where a, b, and c are real numbers. Its graph is a parabola with a vertical axis of symmetry, and its domain is $(-\infty, \infty)$.

Quadratic equation

A **quadratic equation in standard form** is

$$ax^2 + bx + c = 0, \qquad a \neq 0.$$

Its solutions are the zeros of the related quadratic function and the x-coordinates of its x-intercepts.

For $f(x) = 2x^2 + 3x - 2$, identify a, b, and c.

- a, b, c are numbers
- quadratic functions is x^2 after transformation.
- graph ∪

Zero-Product Property

Zero-product property

If A and B are real numbers or algebraic expressions and

$$AB = 0,$$

then $A = 0$ or $B = 0$ (or both).

$A \cdot B$

ex $3 \cdot 0 = 0$

$0 \cdot 0 = 0$

Example. Solve for x.

$$(x + 6)(2x - 3) = 0.$$

Then write the corresponding x-intercepts.

$$\begin{cases} x+6=0 \\ 2x-3=0 \end{cases} \Rightarrow \quad x=-6, \quad x=\frac{3}{2}$$

Zeros by Factoring

1. Write the equation in standard form.
2. Factor completely.
3. Apply the zero-product property.
4. Solve each resulting linear equation.
5. Check in the original equation.

$x^2+5x+6=0$ (b, c)

$x^2+bx+c=0$

$(x+2)(x+3)=0$ (a)

$\Rightarrow x=-2,-3$

multiply $= c$, 6; sum $= b$, 5

Example. Find the zeros of

$$f(x) = 2x^2 + 3x - 2.$$

Solve for x when $f(x)=0$.

$2x^2+3x-2=0$

$a=2,\ b=3,\ c=-2$

2) • find $a\cdot c=-4$

• find 2 numbers, "x" $=-4$ ($a\cdot c$), "$+$" $=3$ (b).

so 4, -1 will satisfy.

• rewrite this equation,

$2x^2+3x-2=0$

$2x^2+4x-2x-2=0$

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iii) • rearrange them

• factor

$(-1x-2)+2x^2+4x=0$

$x=2,\ \frac{1}{2}$

$-2(x+2)+2x(x+2)=0$

$(x+2)\cdot(2x-1)=0$

Factoring Patterns

$(a^2-b^2)=(a+b)(a-b)$

Solve each equation by factoring. Name the pattern or common factor you use.

1. Common factor

$4x^2 - 20x = 0$

$\frac{4x^2}{4x} = x \quad \frac{-20x}{4x}$

$\Rightarrow$

$4x(x-5)=0$

$a=2, b=5, c=-7$

2. Difference of squares

$3x^2 - 108 = 0$

$3(x^2-6^2)=0$

$3(x+6)(x-6)=0$

3. Trinomial: $2x^2 + 5x - 7 = 0.$

i) $a \cdot c = -14$

"x" $=-14$, "+" $=5$ $\Rightarrow 7, -2$

ii) $2x^2+7x-2x-7=0$

$-2x-7+2x^2+7x=0$

$-1(2x+7)+x(2x+7)=0$

$(2x+7)\cdot(x-1)=0$

reminder: $\sqrt{4} = 2$

Square-Root Property

$x^2 = 4$ $\quad \sqrt{x^2} = \pm\sqrt{4}$

$x = \pm 2$

Square-root property

If u is an algebraic expression and $d > 0$, then

$$u^2 = d \implies u = \pm\sqrt{d}.$$

The symbol $\pm$ is necessary because both a positive and a negative number can have the same square.

Solve each equation using the square-root property.

$x^2 - 12 = 0$

$\sqrt{x^2} = \sqrt{12}$

$x = \pm\sqrt{12}$

$x = \pm\sqrt{4 \cdot 3} = \pm\sqrt{4}\cdot\sqrt{3}$

$= \pm 2\sqrt{3}$

$(x-2)^2 - 16 = 0$

$(x-2)^2 = 16$

$x - 2 = \pm\sqrt{16}$

$x - 2 = \pm 4$

$x = 6, -2$

$\sqrt{3+2} \neq \sqrt{3} + \sqrt{2}$

$\sqrt{3}\cdot\sqrt{2} = \sqrt{3}\cdot\sqrt{2}$

$x^2+2xy+y^2=(x+y)^2$ $x^2-2xy+y^2=(x-y)^2$

Completing the Square

For a binomial $x^2 + bx$, add the square of half the coefficient of x:

$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$$

Complete each expression so that it becomes a perfect-square trinomial. Then factor it.

$b=8$

$x^2 + 8x + \left(\frac{8}{2}\right)^2$

$= x^2 + 8x + 4^2$

$= (x+4)^2$

$b=-14$

$x^2 - 14x + \left(\frac{-14}{2}\right)^2$

$= x^2 - 14x + 7^2$

$= (x-7)^2$

Solve by Completing the Square

To solve $ax^2 + bx + c = 0$:

1. Divide by a if $a \neq 1$.
2. Move the constant term to the other side.
3. Add $\left(\frac{b}{2}\right)^2$ to both sides.
4. Factor the perfect-square trinomial.
5. Apply the square-root property and simplify.

Example. Solve $x^2 + 8x + 10 = 0$.

$x^2+8x = -10$

Observe $b=8$,

$x^2+8x+\left(\frac{8}{2}\right)^2 = -10+\left(\frac{8}{2}\right)^2$

$x^2+8x+4^2 = 6$

$(x+4)^2 = 6$

$x+4 = \pm\sqrt{6}$

$x = -4 \pm \sqrt{6}$

$x=-4+\sqrt{6}$ or $x=-4-\sqrt{6}$

Method Practice

Choose factoring, the square-root property, or completing the square. State why your method fits.

1. $x^2 + 5x - 1 = 0$
2. $2x^2 - 4x - 5 = 0$
3. $(3x + 1)^2 = 20$

1) $x^2+5x=1$

$x^2+5x+\left(\frac{5}{2}\right)^2=1+\left(\frac{5}{2}\right)^2$

$\left(x+\frac{5}{2}\right)^2=\frac{29}{4}$

$x+\frac{5}{2}=\pm\sqrt{\frac{29}{4}}$

$x=-\frac{5}{2}\pm\sqrt{\frac{29}{4}}$

$\sqrt{\frac{a}{b}}=\frac{\sqrt{a}}{\sqrt{b}}$

$=-\frac{5}{2}\pm\frac{\sqrt{29}}{2}=\frac{-5\pm\sqrt{29}}{2}$

2) $\frac{2x^2-4x}{2}=\frac{5}{2}$

$x^2-2x=\frac{5}{2}$, $b=-2$

$x^2-2x+\left(\frac{-2}{2}\right)^2=\frac{5}{2}+\left(\frac{-2}{2}\right)^2$

$x^2-2x+1^2=\frac{7}{2}$

$(x-1)^2=\frac{7}{2}$

$x-1=\pm\sqrt{\frac{7}{2}}$

$x=1\pm\sqrt{\frac{7}{2}}$

3) $(3x+1)^2=20$

$3x+1=\pm\sqrt{20}$

$3x=1\pm\sqrt{20}$

$x=\frac{1\pm\sqrt{20}}{3}$